Project Planning & Work Breakdown Structure

Date	18 October 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

1. Project Milestones & Timeline (12-Week Roadmap)

Milestone	Phase / Activity	Duration	Deliverables	Status
M1	Problem Definition & Requirements	Weeks 1 - 2	Problem statements, customer journey map, DFD, technology stack.	**Completed**
M2	Computer Vision & 3D Spatial Canvas	Weeks 3 - 4	RT-DETR object detection, depth estimation, Three.js 3D canvas viewport.	**Completed**
M3	Audio Intelligence & Transcription	Weeks 5 - 6	Whisper ASR integration, acoustic classifier, 4-language translation tabs.	**Completed**
M4	CCTV Video Temporal Timeline	Weeks 7 - 8	1.0 Hz video decomposition, BLIP captioning, frame-by-frame timeline.	**Completed**
M5	Multimodal Fusion & Groq Reasoning	Weeks 9 - 10	Cross-modal timestamp synchronization, Groq LLaMA-3.3 forensic chatbot.	**Completed**
M6	Court PDF Reporting & Testing	Weeks 11 - 12	ReportLab PDF generator, unit test suite (5/5 passing), external data validation.	**Completed**

2. Risk Management & Mitigation Matrix

Risk Description	Severity	Likelihood	Mitigation Strategy Implemented
Large ML models failing on CPU hardware.	High	Medium	Dual-tier lazy loading with local cache fallback and lightweight CV heuristic detectors.
Audio transcripts containing XML characters crashing PDF.	High	High	Applied xml.sax.saxutils.escape sanitation across all ReportLab Paragraph injections.
Three.js canvas crash when switching hidden tabs.	Medium	Medium	Added zero-dimension guards inside onResize() to prevent NaN camera aspect ratios.